

Superstore Sales Analysis – Business Insights & Reporting

1. Dataset Overview

The Sample Superstore dataset contains sales transaction records from a retail business. The dataset includes information related to customers, products, sales, profits, discounts, shipping details, and regions.

Key Columns:

- Order ID
- Order Date
- Ship Date
- Customer Name
- Segment
- Region
- Category
- Sub-Category
- Sales
- Profit
- Discount

The dataset was used to analyze sales performance, customer behavior, product performance, and business profitability.

2. Data Cleaning Summary

Before analysis, the dataset was cleaned and prepared using Microsoft Excel.

Data Cleaning Steps:

- Checked for missing values.
- No missing values were found.
- Checked for duplicate records.
- No duplicate records were found.
- Standardized date formats.
- Created Profit Margin column.
- Created Order Processing Days column.

- Created Month column for trend analysis.

After cleaning, the dataset was ready for SQL analysis, dashboard creation, and exploratory data analysis.

3. SQL Findings

Several SQL queries were executed to analyze business performance.

Key findings include:

- Total Sales and Total Profit were calculated successfully.
- Top 10 customers contributed a significant portion of total revenue.
- Sales performance varied across regions.
- Certain categories were more profitable than others.
- Customer purchase behavior was analyzed using order frequency and spending patterns.
- Total Orders and Total Quantity Sold were calculated successfully.
- Sub-Category wise profit can be calculated successfully.
- Profit by segment and Average discount by category can be find.

4. Dashboard Insights (Power BI)

A multi-page interactive dashboard was created in Power BI.

Executive Summary:

- Total Sales
- Total Profit
- Total Orders
- Average Order Value

Sales Analysis:

- Region-wise Sales
- Category-wise Sales
- Monthly Sales Trend

Customer Insights:

- Top Customers
- Customer Segments

Product Performance:

- Best Selling Products
- Most Profitable Products

Dashboard Observations:

- Sales varied across regions.
- Technology products generated strong revenue.
- A small number of customers contributed significantly to total sales.
- Monthly sales showed seasonal fluctuations.
- Product profitability varied across categories.

5. Python EDA Findings

Exploratory Data Analysis was performed using Python libraries such as Pandas and Matplotlib.

Analysis Performed:

- Dataset Overview
- Missing Value Analysis
- Sales Trend Analysis
- Profit Trend Analysis
- Product Analysis
- Customer Analysis

Visualizations Created:

1. Monthly Sales Trend
2. Monthly Profit Trend
3. Top 10 Products by Sales
4. Top 10 Customers by Sales
5. Category-wise Profit
6. Region-wise Sales
7. Region-wise Profit
8. Discount vs Profit

Key EDA Findings:

- Technology category generated the highest revenue.

- High discounts negatively impacted profitability.
- Certain regions consistently outperformed others.
- Top customers contributed a large portion of revenue.
- A few products generated most of the sales volume.

6. Top 5 Business Insights

- Technology products generated the highest sales and profit among all categories.
- A small group of customers contributed a significant share of total revenue.
- Sales performance varied considerably across different regions.
- Higher discounts often resulted in lower profits and reduced margins.
- Certain products consistently appeared among the top-selling and most profitable items.

7. Top 5 Recommendations for Management

- Increase marketing efforts for high-profit categories such as Technology.
- Review discount strategies to avoid unnecessary profit reduction.
- Implement customer loyalty programs for top-performing customers.
- Promote best-selling products through targeted marketing campaigns.
- Develop region-specific sales strategies to improve performance in low-sales regions.

Conclusion

The Superstore Sales Analysis project successfully combined Excel, SQL, Power BI, and Python to generate actionable business insights. The analysis identified key sales drivers, customer trends, product performance, and profitability patterns. These insights can help management make informed decisions to improve sales performance, customer retention, and overall business growth.